

Case Study: Virtual Battery Auction Pricing

Context

Terralayr offers virtual battery capacity to customers via competitive **Virtual Battery Auctions (vBAs)**.

Each auction grants the right to optimize a virtualized slice of a Battery Energy Storage System (BESS) across specified wholesale electricity markets.

These virtual batteries are typically backed by Terralayr-owned physical assets. If an auction does not clear above Terralayr's internal hurdle price, the asset capacity is instead allocated to internal optimizers who trade the battery in the market.

Your task is to estimate the **expected market value of the battery during the auction period** and recommend an appropriate **minimum clearing price (hurdle rate)** for Terralayr.

The objective of this exercise is to demonstrate how you would build a **transparent and reproducible forecasting framework** to estimate BESS wholesale revenues using historical market data, forward-looking price forecasts, and reasonable modelling assumptions.

1. Auction Details

a. Auction Product

You should evaluate hurdle rates for three possible virtual battery auction durations:

- **1 day**
- **1 week**
- **1 month**

Auction Timing

For the purpose of this exercise assume:

- Auctions close **two days before delivery (i.e., D-2) at 17:00 CET**
- Successful bidders can therefore participate in all relevant **markets that have gate closure after D-2 at 17:00 CET** for the delivery period.

- Current date (auction closing): 14 March 2026 (D-2)
 - Delivery start: 16 March 2026 (D)
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b. Auction Pricing Structure

Virtual Battery Auctions are priced as a **fixed payment per MW of capacity** for the full auction period.

The winning bidder pays:

€/MW for the full auction duration

In return, the bidder receives the right to optimize the battery across the specified markets and retains all realized trading revenues.

Terralayr should only award capacity if the auction clearing price **exceeds the internally estimated value** of operating the battery during the same period.

Your task is therefore to estimate the **expected revenue potential of the battery** during the auction period and determine the **minimum acceptable auction clearing price (€/MW)** Terralayr should accept.

2. Asset and Market Data

a. Asset Parameters

Virtual battery characteristics:

- Power capacity: **10 MW**
- Energy capacity: **20 MWh**
- Charge efficiency: **95%**
- Discharge efficiency: **95%**
- Maximum cycles per day: **2**

Operational assumptions:

- Fully available during the auction period
- No degradation costs
- No ancillary service participation

- Optimization allowed in **wholesale markets only**

Available markets:

- **Day-Ahead (DA) auction**
 - **Intraday continuous (IDC)**
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b. Market Data

Historical market data from **01 January 2025 onwards** is provided for:

- EPEX Day-Ahead prices
- Intraday continuous price indices
- Independent **2-hour wholesale-only German BESS revenue index**

Market Design Change

The German Day-Ahead market transitioned from hourly settlement to 15-minute settlement in October 2025.

As a result:

- Historical datasets prior to October 2025 contain hourly DA prices
 - Later datasets contain 15-minute DA prices
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Forward-Looking Market Data

Forecast datasets include:

- Day-Ahead price forecasts
- Renewable generation forecasts
- Electricity demand forecasts

Forecast datasets contain **both recent actual and forecast values** with 15-min granularity.

Fundamentals Data

Fundamental datasets contain:

- Renewable generation

- Electricity demand
- Residual load

Longer historical periods are provided as **daily averages**.

BESS Revenue Benchmark

An independent **2-hour German BESS wholesale revenue index** is provided.

This index is expressed as **annualized revenue (kEUR/MW/year equivalent)** derived from daily market opportunities across all wholesale markets and can be used as a reference benchmark for validating your modelling results.

3. Tasks

a. Battery Revenue Model

Develop a simplified model to estimate the wholesale arbitrage value of the battery.

Your model should reflect:

- power constraints
- energy constraints
- charging and discharging efficiency
- market price signals

The goal is to estimate the revenue potential of the battery under different market conditions.

You may use:

- optimization approaches
- heuristic dispatch rules
- statistical approximations

Clearly explain:

- modelling assumptions
- simplifications
- limitations of your approach.

b. Revenue Forecasting

Develop a methodology to estimate future BESS revenues for the upcoming:

- 1 day
- 1 week
- 1 month

auctions.

You are encouraged to develop your **own forecasting approach**, rather than relying solely on the provided Day-ahead price forecast.

Where possible, demonstrate how your methodology performs using **historical backtesting**.

Provide an estimate of the **expected revenue distribution** for the battery over the relevant auction horizons.

c. Auction Hurdle Price

Based on your revenue forecasts and BESS optimization model, recommend a **minimum auction clearing price (€/MW)** for the full auction duration.

Discuss:

- expected revenues
 - uncertainty and downside risk
 - differences across auction durations.
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d. Model Validation

Compare your estimated historical battery revenues with the **provided BESS revenue index**.

Discuss:

- similarities and differences
- potential reasons for deviations

- limitations of your modelling approach.

Provide **quantitative error metrics** where possible.

e. Conceptual Extension (No Implementation Required)

Describe how you would extend your modelling framework to address the following:

- **Cross-market optimization**, including participation in Intraday Auctions (IDA) and ancillary service markets such as **FCR or aFRR**
- **Revenue forecasting for longer horizons** such as a quarterly product.
- **Assets with Flexible Connection Agreement (FCA) constraints**

Focus on the conceptual modelling challenges, required inputs, and potential modelling approaches.

4. Deliverables

Please submit:

- **5–7 slides/pages** presenting your results
- supporting code

Your slides should clearly communicate:

- modelling approach
 - key assumptions
 - main results
 - recommended hurdle price.
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Assumptions

List clearly and explain:

- modelling assumptions
- how you handled data imperfections
- any approximations used.

5. Evaluation Criteria

Candidates will be assessed on:

- Quality of quantitative reasoning and modelling approach
 - Forecasting methodology and interpretation of market data
 - Practicality of the recommended auction pricing
 - Clarity of communication and structure
 - Visualization and storytelling
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6. Code

Your code should reflect your approach to data analysis and modelling.

Code submissions are welcome in formats such as:

- Python scripts
- Jupyter notebooks
- other formats that facilitate review and understanding.

Ensure your code is **well documented and easy to follow**.

Submission

Please submit your slides and code to the email address provided (xh@trlyr.com, sw@trlyr.com) **at least one day prior to the presentation**.

During the interview you will be asked to **present your results and discuss your modelling approach**.